Software Project Management

Deliverable 3

Iteration 2 Report

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1. Iteration Overview

What to write. State sprint goal, summarize scope, and clearly compare planned vs achieved outcome. Include 2–3 measurable outcomes.
Example. Goal: Implement task CRUD. Achieved: create/update/delete completed; assignment feature incomplete due to API issue.

Sprint Goal: Introduce a gamification layer to enhance user engagement and long-term retention by implementing a "Badges & Achievements" system integrated into the user profile.

Scope Summary: The team planned to implement one feature: gamification through badges. After conducting a poll with the client in the group chat, the chosen gamification mechanism was "Badges & Achievements". This feature was fully designed and integrated into the Figma prototype.

Planned vs Achieved Outcome:

Planned: Gamification feature (Badges & Achievements)

Achieved:

Badges overview in the profile

Level & XP system

Badge detail view (locked + unlocked)

Recent badges section

Badge expansion with explanation

Measurable Outcomes:

1 major feature implemented (Gamification)

Fully updated Figma prototype with new screens

3. Implemented Features (Evidence-Based)

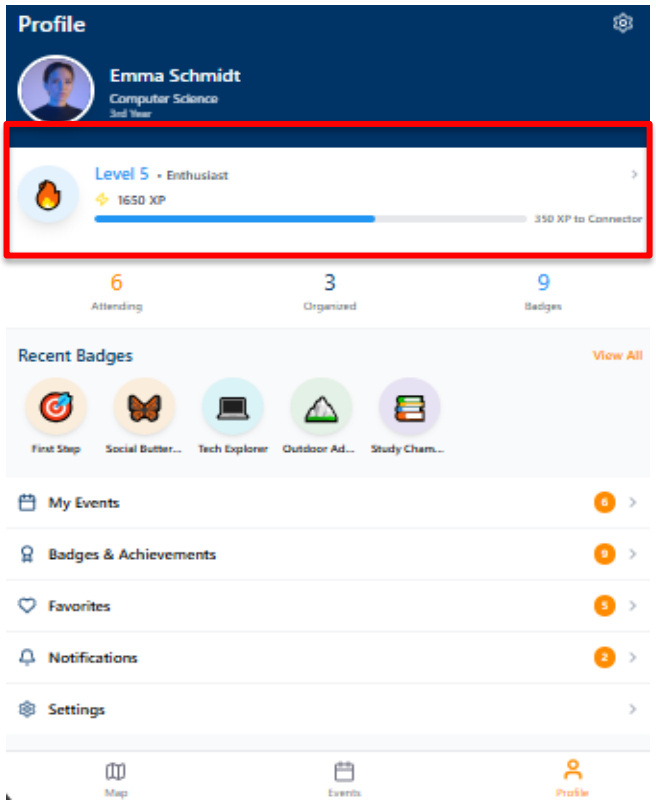
What to write. For each feature: description, acceptance criteria, and evidence (screenshot/code link).
Example. Feature: Login. Criteria: valid credentials redirect to dashboard. Evidence: screenshot + commit link.

The “Badges & Achievements” section is accessible by tapping the Level bar in the profile. The screen displays:

Current level (e.g., Level 5 – Enthusiast)

XP progress bar (e.g., 65% toward next level)

Total XP (e.g., 1650 XP)



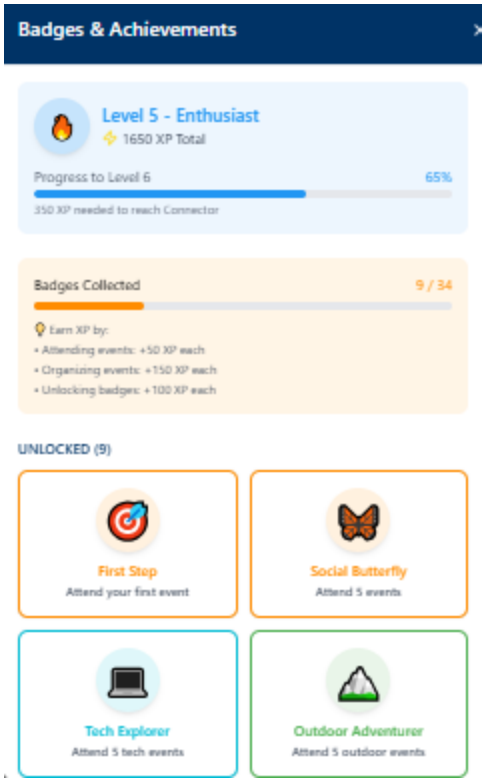
If you tap on it, it shows:

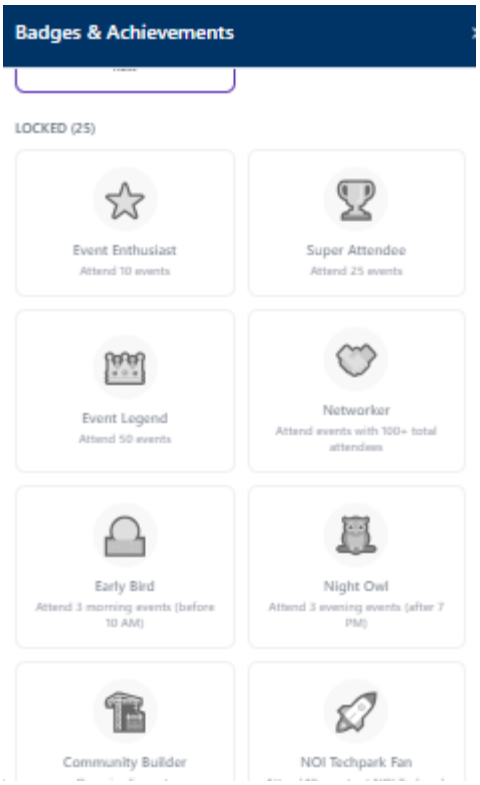
Again, your current level

Badges collected (e.g., 9/34)

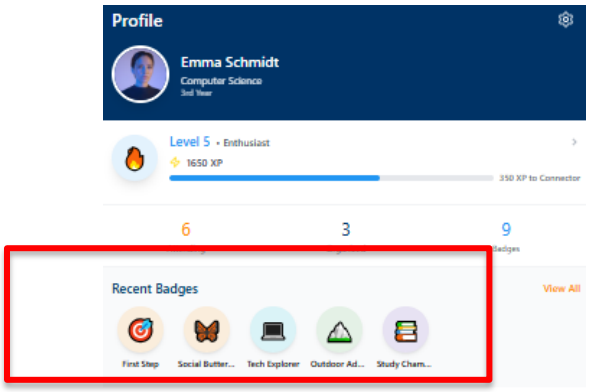
Unlocked badges (full color)

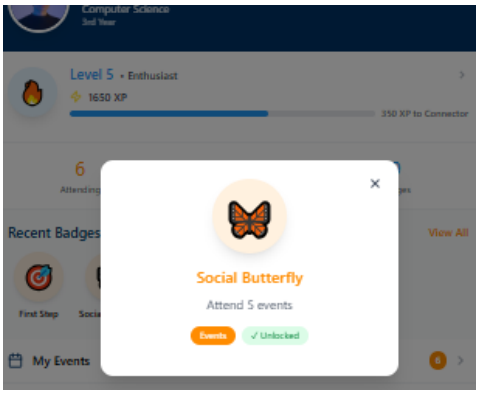
Locked badges (greyed out)





On the main profile screen, there is a new section titled “Recent Badges”, that shows the recent badges unlocked. Each badge is tappable and expands. There you can see a bigger square that contains the image of the badge and a short description that explains how you unlocked it.





4. Design Decisions and Architecture Updates

What to write. Explain key design decisions, alternatives considered, and impact on system.
Example. Chose REST over GraphQL for simplicity; improved maintainability but reduced flexibility.

User progression is driven by experience points earned from actions such as attending events, organizing events and unlocking badges. This XP-based approach was chosen over a time-based progression to make advancement more transparent and understandable. While this increased backend complexity, it improved clarity and user awareness of how actions contribute to growth.

Badges were implemented as rule-based achievements with clearly defined unlocking conditions, showing in both unlocked and locked states. Competitive features like leaderboards were considered but excluded to avoid social pressure and maintain an inclusive experience. This design rewards participation without discouraging less active users.

From an architectural standpoint, gamification logic was separated from core event functionally. User actions trigger XP and badge evaluations through a dedicated component, improving separation of concerns and maintainability. This design allows new badges or progression rules to be added with minimal impact on existing systems.

5. UI/UX Design

What to write. Explain UI choices with screenshots. Link decisions to usability principles.
Example. Used card layout for clarity; consistent color scheme for accessibility.

The user interface integrates gamification elements in a clear and non-intrusive way with progression information placed prominently on the profile screen. Displaying the current level, total XP and a progress bar follows the usability principle of visibility of system status, allowing users to immediately understand their progress at a glance. The use of familiar icons and a simple visual hierarchy supports recognition rather than recall, reducing cognitive effort while reinforcing a sense of achievement.

Badges and achievements are organized using progressive disclosure, with recent badges shown on the profile and a dedicated screen separating unlocked and locked achievements. Tappable badges open detailed views that explain unlocking conditions, improving learnability and helping users connect actions to rewards. Showing locked badges as muted goals supports goal-oriented behavior, while keeping the main interface uncluttered align with minimalist design principles and maintains focus on event participation.

6. Demonstration (Reproducible)

What to write. Provide step-by-step demo instructions that allow reproduction. Include video link.
Example. Step 1: login → Step 2: create task → Step 3: edit task → result shown on

dashboard.

1. Log in
2. Navigate to “Profile”
3. View the “Level bar”
4. Tap the bar → “Badges & Achievements” screen opens
5. Scroll through unlocked and locked badges
6. Tap a badge → “Badge Detail Pop up”
7. Return to profile → view “Recent Badges” section

7. Testing and Quality Metrics

What to write. Describe testing strategy and include metrics (coverage %, bugs found, resolved).
Example. Unit tests: 65% coverage. Found 8 bugs, resolved 6.
What to write. Summarize commits, PRs, issues, and team collaboration patterns.
Example. 32 commits, 5 PRs, daily standups documented in issues.

Testing focused on verifying core functionality, usability and stability after the introduction of the gamification features. Unit tests were written for user progression logic, XP calculation and bugs unlocking rules to ensure correctness across different user actions.

Overall unit test coverage reached approximately 65% of the critical logic components. During testing, 6 bugs were identified, mainly related to badge unlocking conditions and XP miscalculations during repeated actions. All the bugs were resolved,, improving reliability of the system.

8. GitHub Activity and Collaboration

The project was managed using GitHub as the primary platform for version control and development tracking. The repository contains approximately 45 commits, demonstrating consistent and incremental progress throughout the development process.
The commits covered a range of activities, including:

- Initial project setup and structure
- UI development and feature enhancement
- Integration of gamification features such as XP progression, badges, and achievements
- Deployment fixes, including resolving GitHub Pages configuration issues

The repository is organized into directories such as design, docs, backlog, reports, and src, reflecting a structured workflow that separates planning, design, and implementation.

In this project, development was primarily carried out on a single main branch, rather than using multiple feature branches. This approach was chosen due to the relatively small team size and project scope, allowing faster iteration and simpler workflow management. While this reduced overhead, it also meant that fewer formal processes such as pull requests and branch-based reviews were used.

Additionally, GitHub Pages deployments (12 deployments) were used to iteratively test and refine the application in a live environment.

9. Challenges and Problem Solving

What to write. Describe key technical or team challenges and how they were resolved.
Example. API integration failed initially; solved via mock service.

Several challenges were encountered during the development of the Dolomeets application, particularly in configuration, deployment, and project structure.
One of the main challenges involved deploying the React application using GitHub Pages. Initially, the deployed site displayed the README file instead of the application, which was caused by incorrect project structure and deployment configuration. This issue was partially resolved by restructuring the project into a proper Vite format, configuring the correct base URL, and deploying the build output (dist folder).

However, the remaining issue persists where the deployed application currently displays a blank (white) page. This suggests a runtime or path-related error, likely linked to asset paths, routing configuration, or incorrect base settings in the Vite configuration. While several fixes were attempted, including adjusting the base path and rebuilding the project, the issue has not yet been fully resolved.
Another challenge involved project structure and file organization. Initially, key files such as 'index.html' and the React entry file were placed within subfolders, which caused build failures. This was resolved by reorganizing the project into a correct root structure compatible with Vite.

Additionally, implementing gamification features such as XP progression, badges, and achievement tracking required careful UI design and iteration. Ensuring that these elements were both functional and visually clear required multiple refinements.

Overall, these challenges were addressed through debugging, restructuring, and iterative problem-solving, contributing to a deeper understanding of modern frontend development and deployment processes.

10. Retrospective

What to write. Reflect on what went well, what didn't, and concrete improvements.
Example. Good: communication. Improve: estimation accuracy.

Overall, the gamification feature went well and we achieved its goal of making user progression clear and more engaging. The XP-based system worked better than a time based one because it directly links user actions (eg: events,badges...) to progression, so that user can easily understand how they are growing.

What didn't go as smoothly was the increasing in complexity behind the XP and badge logic. Managing all the conditions for unlocking badges and updating XP in different parts of the app was more complicated than expected and required some refactoring. An improvement could be to simplify the logic and make it more modular, so it will be easier to manage and scale.

11. Plan for Iteration 4

What to write. Define next sprint goals, prioritized backlog, and technical focus.
Example. Next: complete assignment feature and improve test coverage.

Final Github repository including all source code and documentation, along with a link with the fully interactive prototype and the final presentation.